Project 04 — Professor Q&A Follow-Up

Predictive Maintenance for Industrial Equipment (NASA CMAPSS FD001)

STAT 1100 · Spring 2026 · Jongmin Lee & Aedrian Formento · March 31, 2026

Q1. Why does RUL exceed 350 in the PCA visualization?

Slide 4 displays the PCA-projected sensor space colored by **raw (uncapped) RUL**. In the CMAPSS FD001 dataset, engine lifespans range from 128 to 362 operating cycles. An engine at cycle 1 with a total lifespan of 362 naturally has RUL = 361 — this is the source of values exceeding 350.

In short: RUL = max_cycle − current_cycle. Since no cap is applied at the EDA stage, values naturally reach up to 362.

Why show raw RUL instead of capped?

The EDA slide intentionally uses raw RUL to reveal the **natural data distribution** before any preprocessing. This makes two things visible:

1. The wide lifespan variance across the 100-engine fleet (128–362 cycles).
2. The fact that high-RUL points (healthy engines) form a dense, indistinguishable cluster in sensor space — motivating the RUL cap at T=130 in the modeling stage.

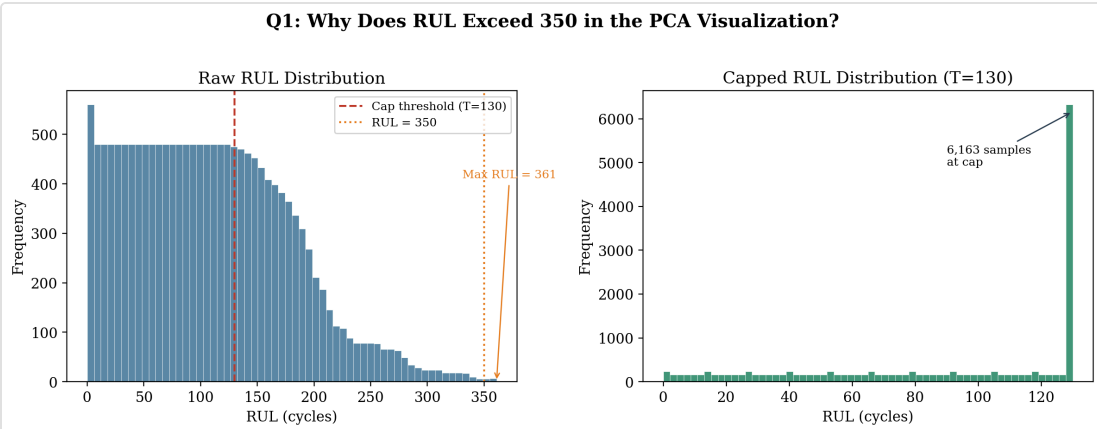

Figure 1. Raw RUL distribution (left) showing values up to 361, vs. capped distribution (right) with 7,633 samples pinned at T=130.

> **Key point:** The RUL cap (T=130) is a *modeling* decision, not a data-cleaning step. Slide 4 correctly shows the pre-cap view because its purpose is exploratory — showing how sensor-space structure relates to the full range of engine health.

Q2. Why was only T ≤ 30 used for failure classification? What about T > 30?

The binary classifier on Slide 12 uses a 30-cycle failure horizon: *"Will this engine fail within 30 cycles?"* This threshold was selected based on three considerations (documented in ADR-005):

Original rationale for T=30

1. **Class balance.** T=30 yields ~15% positive class — imbalanced but manageable without resampling techniques like SMOTE.
2. **Operational lead time.** 30 cycles provides sufficient advance notice for maintenance scheduling in turbofan operations.
3. **Sensor detectability.** Within 30 cycles of failure, sensor degradation is clearly measurable, giving the classifier reliable signal.

Extended analysis: T=10 through T=100

We ran the same KNN classifier (K=19) across eight threshold values. All models use identical features and preprocessing; only the binary label changes.

Threshold	F1 Score	Precision	Recall	Positive Rate
10	0.850	0.911	0.795	5.3%
15	0.848	0.901	0.800	7.7%
20	0.877	0.900	0.855	10.1%
30	**0.860**	**0.895**	**0.827**	**15.0%**
40	0.878	0.912	0.846	19.8%
50	0.862	0.925	0.807	24.6%
70	0.854	0.939	0.783	34.3%
100	0.827	0.908	0.758	48.8%

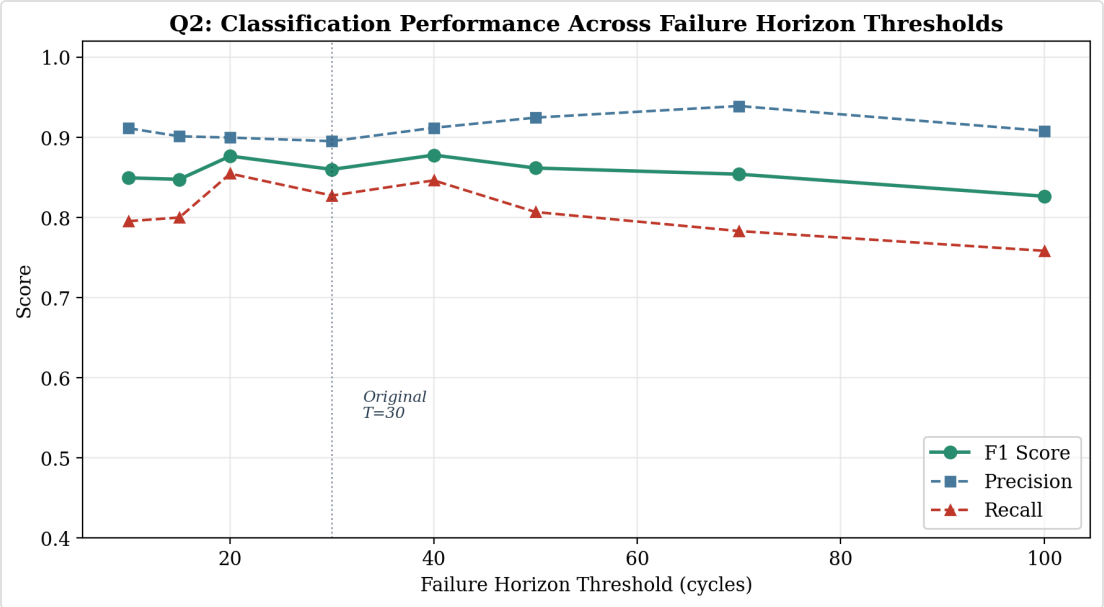

Figure 2. F1, Precision, and Recall across failure horizon thresholds (T=10–100). Performance is stable in the 20–50 range.

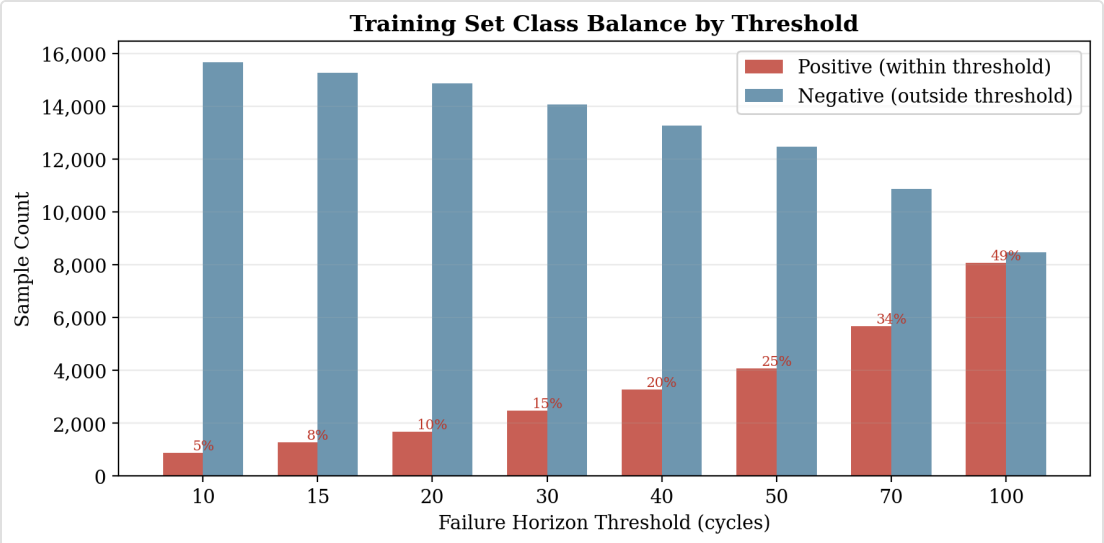

Figure 3. Training set class balance shifts as the threshold increases. At T=100, classes approach 50/50.

Observations

- **F1 is remarkably stable** across T=20–50 (0.86–0.88), indicating the model generalizes well across different alert windows.
- **Precision increases with larger T** (peaking at 0.939 for T=70) because more samples fall into the positive class, reducing false positives.
- **Recall decreases with larger T** (dropping to 0.758 at T=100) because the classifier must detect subtler, earlier degradation signals.

- **T=30 is a sweet spot** for the business use case: it maintains F1 ≥ 0.86 while providing an actionable 30-cycle maintenance window — long enough to schedule but short enough for sensor signals to be strong.

As the threshold increases, precision rises but recall drops. T=30 strikes the practical balance between detection rate and actionability.

> **Key point:** Higher thresholds (T>30) are *classifiable* but less *actionable*. At T=70 or T=100, the model would trigger alerts so early that sensors have barely begun to degrade, leading to either premature maintenance or alert fatigue.

Q3. Why cluster into K=3 instead of the statistically optimal K=2?

The silhouette analysis identified K=2 as the statistically optimal number of clusters (silhouette score = 0.430 vs. 0.334 for K=3). Despite this, we chose K=3 for the following reasons:

Domain-driven interpretability

Turbofan engine degradation follows a well-established three-phase lifecycle:

Cluster	Interpretation	Samples	Mean RUL	Median RUL
0	Healthy	6,271	155.3	148.0
1	Degrading	7,229	104.5	95.0
2	Critical	3,061	21.4	19.0

K=2 merges the *degrading* and *critical* phases into a single cluster, losing the operationally important distinction between "beginning to wear" and "approaching failure."

K=2 is statistically optimal, but K=3 maps directly to the three-phase engine lifecycle (healthy / degrading / critical), making it far more interpretable.

Independent validation

When the K=3 cluster assignments are compared against actual RUL values, the unsupervised clusters track the supervised RUL gradient closely. This alignment confirms that the three degradation stages are **real structure in the data**, not artifacts of the labeling scheme.

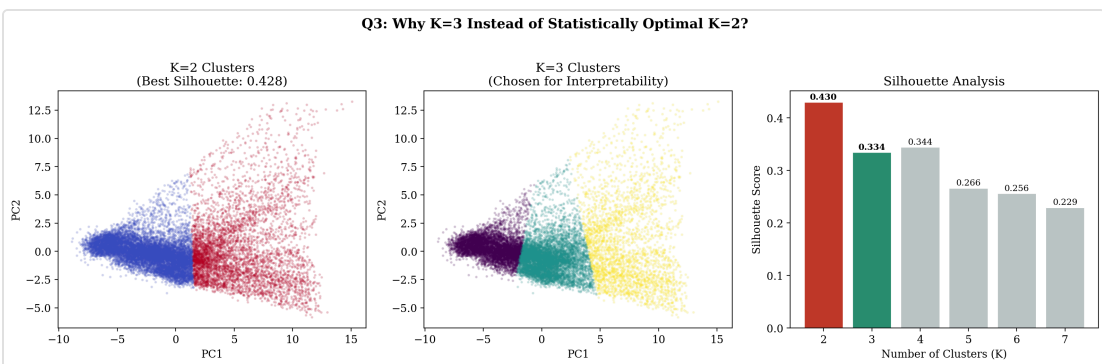

Figure 4. K=2 (left) vs. K=3 (center) in PCA space, with silhouette scores (right). K=3 captures the tri-phase degradation pattern despite lower silhouette.

Why the silhouette gap is acceptable

- The silhouette difference is modest (0.430 vs. 0.334 — both below 0.5, indicating overlapping clusters regardless).
- In degradation processes, cluster boundaries are *inherently gradual*; sharp separation is not expected.
- K=3 aligns with the three-regime structure visible in individual sensor time-series plots (stable → drifting → accelerating).

> **Key point:** Silhouette score optimizes for geometric compactness. In predictive maintenance, *interpretability and domain alignment* take priority — plant managers need to act on "healthy / degrading / critical," not "cluster A / cluster B."